

Shafiq-us Saleheen

Computer Science PhD Student | Purdue University

Email: ssalehee@purdue.edu · Cellphone: +1 765 543 0449

Portfolio: <https://saleheenshafiq9.github.io/>

RESEARCH INTERESTS

Autonomous systems security, software vulnerability discovery, and program analysis (including fuzzing and binary/patch analysis).

EDUCATIONAL BACKGROUND

Purdue University - West Lafayette, Indiana, USA

PhD in Computer Science

Aug 2025

- PurSec Lab (Advised by Dr. Antonio Bianchi)

Institute of Information Technology, University of Dhaka, Bangladesh

B.Sc. in Software Engineering

Jan 2019 - Dec 2023

- CGPA: 3.86/4.00

IELTS

- Overall: 8
 - Listening: 8.5, Reading: 8.5, Speaking: 7, Writing: 7.5
-

Jan 2025

EXPERIENCE

Graduate Research Assistant

Purdue University, West Lafayette, IN, USA

Jan 2026 - Present

- Currently working as an RA with Dr. Antonio Bianchi & Dr. Dongyan Xu
 - Working on fuzzing frameworks especially for unmanned autonomous vehicles (UAVs)
 - Investigating on different forms of binary and patch analysis techniques
 - Developing time-sensitive architectures for Drone software

Graduate Teaching Assistant

Purdue University, West Lafayette, IN, USA

Aug 2025 - Dec 2025

- Worked as a GTA for CS251: *Data Structures & Algorithms* -
 - Leading a PSO (Practice Study Observation) session
 - Holding Office Hours
 - Involved in grading, proctoring, question-setting and proof-reading.
- **Technical Skill:** Data Structure & Algorithm basics.
- **Soft Skill:** Effective communication, Teaching, Time management, Teamwork

Software Engineer

Samsung R&D Institute Bangladesh, Dhaka, Bangladesh

Feb 2024 - Jul 2025

- Samsung's SmartThings Web project -
 - Contributed as full-stack developer in recent releases.
 - Packaged the application into containers, deployed and orchestrated them.
- Research & Innovation -
 - Submitted DOI on AR/VR related solutions.
 - Built PoCs for generative AI, UWB/BLE & AR/VR related solutions.

- Training on system design fundamentals, clean code & refactoring, generative AI.
- **Technical Skill:** NEXT JS, Micronaut (Kotlin), DynamoDB, Android SDK, Java, FastApi, Docker, MongoDB, MySQL, Ubuntu Linux, Git, Visualization
- **Soft Skill:** Effective communication, Task adaptation, Time management, Teamwork, Agile principles

Researcher

Governance, Privacy and Security Research Lab (GPSLab)
Pennsylvania State University, University Park, PA, USA

Feb 2025 - Present

- Currently working under the supervision of Dr. Tanusree Sharma

Research Intern

School of Computing and Augmented Intelligence (SCAI)
Arizona State University, Tempe, AZ, USA

Dec 2024 - Present

- Currently working under the supervision of Dr. Ozgur Ozmen

Research Intern

Secure Platforms Lab (SPL)
William & Mary, Williamsburg, VA, USA

Jun 2023 - Dec 2023

- Worked under the supervision of Dr. Adwait Nadkarni

Software Engineering Intern

Streams Tech Ltd.

Dec 2022 - Jun 2023

- Automated data cleaning processes, used chunking & scheduling for Kobo Toolbox R&D.
- Developed an automated error generation tool for conducting checks on shapefiles.
- **Technical Skill:** C#, Python, Docker, PostgreSQL, QGIS, PostGIS

RESEARCH PROJECTS

ICS Log Protection via Integrity Validation

Arizona State University, Tempe, AZ, USA

Dec 2024 - Present

- Collaborated with Dr. Ozgur Ozmen from Arizona State University
- Collaborated with Dr. Antonio Bianchi from Purdue University
- Developing a system for ICS log protection via integrity validation
- **Technical Skill:** ICS simulation tools, Python, GraphSAGE, Transformers, SQLite

RedactoRestore: User Centered Data Minimization Tool in Managing PII in Contemporary Web Editing Tools

Pennsylvania State University, University Park, PA, USA

Feb 2025 - Present

- Collaborated with Dr. Tanusree Sharma from Pennsylvania State University
- Developing a tool for data minimization with redaction
- **Technical Skill:** JavaScript, Python, PII Models
- **Soft Skill:** Technical Writing, Effective communication

"Where's the Line Between Evolution and Exploitation?" - Generative AI & the Postcolonial Evolution of Software Landscape in the Global South

University of Dhaka, Dhaka, Bangladesh

May 2024 - Present

- Conducted a qualitative study on the adoption and impact of Generative AI (GAI) tools in Bangladesh's software industry.
- Integrated post-colonial computing principles to assess the implications for the Global South.
- Analyzed GAI's effects on software processes and highlighted potential job shifts.
- Addressed ethical issues like data dispossession and privacy, advocating for fair AI practices.
- **Technical Skill:** Ethnographic research, Postcolonial computing, Thematic analysis, SDLC
- **Soft Skill:** Open-ended interviews, FGD conduction, Effective communication

IoTWhiz: An IoT Android App Characterization Tool

University of Dhaka, Dhaka, Bangladesh

Jun 2023 - Dec 2023

(In collab. with William & Mary, Williamsburg, VA, United States)

- Developed a tool to characterize IoT/non-IoT android applications using program slicing techniques, providing insights into their structure, behavior, and dependencies.
- Automated report generation, comparing IoT and non-IoT apps across various metrics like code complexity, reflection usage, data storage strategies, and permissions.
- Developed a web interface to control the navigation and show the interpretation of analysis.
- **Technical Skill:** Java, React JS, FastApi, SQLite, JadX, Hypothesis testing, Visualization
- **Soft Skill:** Clean code, Critical thinking, Report writing
- **GitHub:** <https://github.com/saleheenshafi9/IoTWhiz>
- **Report Link:** [PDF](#)

PharmaSync: Optimizing Security and Transparency in Pharmaceutical Supply Chains using Decentralized Hybrid Blockchain

International Blockchain Olympiad 2023, Amsterdam, Netherlands

Jul 2023 - Present

- Developed PharmaSync, to address critical issues in pharmaceutical supply chains, including privacy, drug counterfeiting, and compliance issues.
- Integrated zero-knowledge proofs to enhance security while maintaining transparency.
- Implemented hybrid blockchain architecture to balance transparency and privacy.
- **Technical Skill:** NEXT JS, Hedera consensus, Hyperledger indy, Solidity, Azure, IPFS
- **Soft Skill:** Team work, Problem solving, Technical writing, Presentation skills
- **Paper Link:** [PDF](#)

Cricket Winner Prediction - Domain Based Feature Engineering and Analysis

University of Dhaka, Dhaka, Bangladesh

Sep 2023 - Nov 2023

- The project aims to predict the winner of cricket matches by analyzing internal and external factors of a team. It focuses on feature engineering to improve prediction accuracy.
- It suggested that incorporating additional data, like weather conditions, could improve predictions.
- **Models Used:** Logistic regression, Decision tree, Random forest, and Support vector machine (SVM)
- **Report Link:** [PDF](#)

Software Requirements Specification (SRS) for the ERP System of Chittagong University (CU)

University of Chittagong, Chittagong, Bangladesh

May 2023 - Aug 2023

- Improved the current manual processes of CU by proposing an automated ERP system.
- Made the system more extensible by proposing a modular architecture that supports administrative and academic activities.
- Extended cross-departmental support for all faculties and administrative units within CU.
- **Technical Skill:** Software architecture, Modular design, System automation, Requirements analysis
- **Soft Skill:** Critical thinking, Best practices for documentation workflow

ACADEMIC PROJECTS

Save The Date: A Party Planner

- Developed a party planning web application which serves as a centralized hub, connecting various party service providers (caterers, decorators, vendors etc.) with potential customers.
- Built a recommendation system using K-Nearest Neighbor algorithm to suggest a complete plan for services to avail based on user inputs like date, budget, and location.
- **Technologies:** React JS, Django REST, MongoDB
- **GitHub:** <https://github.com/saleheenshafi9/SaveTheDate>

Chemouflage: An Interactive AR Learning app for Chemistry

- Designed to help students visualize complex chemistry concepts using immersive AR experiences.
 - Displayed molecular structures, chemical reactions and quizzing features using gamification model.
 - **Technologies:** Unity Game Engine, SQLite
 - **Video Link:** <https://www.youtube.com/@Chemouflage>
-

CO-CURRICULAR ACTIVITIES

Online Courses -

- The Art of API Documentation - [Certificate](#) [2024]
 - Python for Computer Vision with OpenCV and Deep Learning - [Certificate](#) [2024]
 - Google Data Analytics Specialization - [Certificate](#) [2021]
 - Google UX Design Specialization - [Certificate](#) [2021]
 - Technical Support Fundamentals - Google - [Certificate](#) [2020]
-

ACHIEVEMENTS

Award of Distinction, 6th International Blockchain Olympiad 2023, Amsterdam, Netherlands

- Won the award for a supply chain solution named - “PharmaSync: Optimizing Security and Transparency in Pharmaceutical Supply Chains using Decentralized Hybrid Blockchain”
- Certificate Link: [PDF](#)

Professor Jamilur Reza Choudhury Championship Award, 4th National Blockchain Olympiad 2023, Dhaka, Bangladesh

- News Link: [‘Team Pharmars’ -the Best team at the 4th Blockchain Olympiad](#)

Finalist, Learnathon 2022, Dhaka, Bangladesh

- Won the award for a twitter clone website using Angular & .NET framework

Finalist, Hult Prize 2023-24 on Campus Program of ULAB, Dhaka, Bangladesh

- Won the award for an idea called “Turning Invasive Suckerfish into Sustainable Fertilizer”.

Champion, RUET CSE Fest 2k22, Rajshahi, Bangladesh

- Won the award for an app for chemistry learning called “Chemouflage” using Unity game engine.
-

SKILLS

- Programming Languages: C, C++, C#, Python, Java, Javascript, Kotlin, R, PHP, Dart, Solidity
 - Front-end Technologies: React JS, Next JS, Angular, Tailwind CSS, Bootstrap, HTML, Flutter
 - Back-end Technologies: FastApi, Django REST, Node JS, Express JS, Micronaut
 - Database Technologies: MySQL, PostgreSQL, SQLite, DynamoDB, MongoDB
 - Others: Docker, Git, Figma, Postman, Design Patterns, Microservice basics, QGIS, Tableau
 - Soft Skills: Leadership, Planning, Presentation, Taking interviews, Teamwork, Documentation, Analytical thinking, Time management
-